

PROJECT MANAGEMENT 3

TERM 4 TEST REVIEW

Things to Note

- Theoretical questions – All chapters
- Scenario based calculations
 - NPV
 - WBS
 - EVM
 - RISK ASSESSMENT
 - EMV
 - CONTINGENCY FUND

The Net Present Value (NPV) Formula

- The formula for the calculation of the net present value can be simplified as

$$PV = \frac{FV}{(1 + r)^n}$$

where:

NPV = Net Present Value

FV = Net Cash Flow of a period

r = Discount Rate or Interest Rate

n = Total current Period (base of 0)

- By calculating this value for each period, we can then add all the values to obtain the NPV.

NPV Calculation

NET PRESENT VALUE (Option 1)

	Year 0	Year 1	Year 2	Year 3	Year 4
Anticipated Savings					
Project Cost					
Future Value					
Present Value					
Net Present Value					

$$PV = FV / (1 + r)^n$$

NET PRESENT VALUE (Option 2)

	Year 0	Year 1	Year 2	Year 3	Year 4
Anticipated Savings					
Project Cost					
Future Value					
Present Value					
Net Present Value					

WBS

Activity		Duration	Earliest		Latest		Slack
No	Description		Start	Finish	Start	Finish	
1	Collect Data	3	0	5	-3	2	-3
2	Study Feasibility	4	0	4	-2	2	-2
3	Project Definition Report	1	5	6	2	3	-3
4	Interview Users	5	6	12	7	13	1
5	Study Existing System	8	6	16	3	13	-3
6	Define User Requirements	5	16	22	13	19	-3
7	Systems Analysis Report	1	22	23	19	20	-3
8	Input & Output	8	23	33	30	30	-3
9	Processing & Database	10	23	27	32	36	9
10	Evaluation	2	33	39	30	36	-3
11	Systems Design Report	2	39	41	36	38	-3
12	Software Development	15	41	61	38	58	-3
13	Hardware Development	12	41	46	58	63	17
14	Network Development	8	41	53	51	63	10
15	Systems Development Report	2	46	51	63	68	17
16	Software Testing	10	53	58	63	68	10
17	Hardware Testing	6	61	71	58	68	-3
18	Network Testing	4	71	77	68	74	-3
19	Testing Report	5	77	82	79	84	2
20	Training	6	77	87	74	84	-3
21	System Conversion	10	87	93	84	90	-3

Packaging machine project develop cumulative budgeted cost

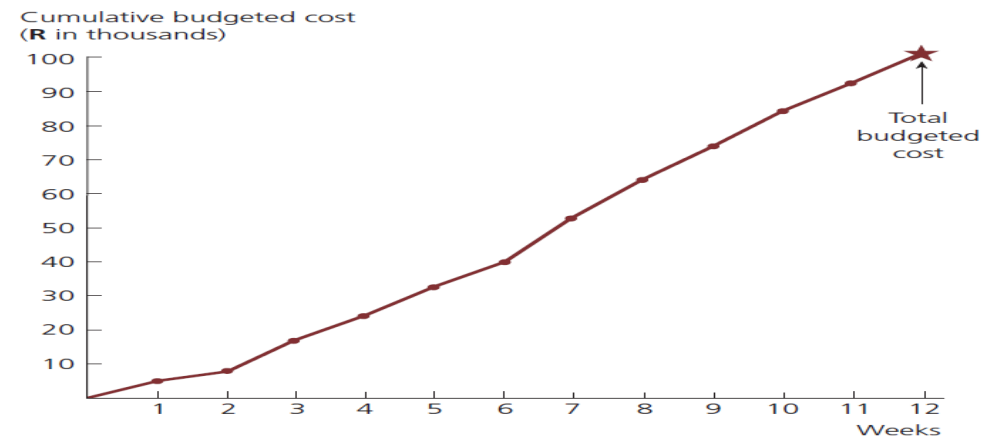
FIGURE 7.5 Budgeted Cost by Period for the Packaging Machine Project

		Week											
	TBC	1	2	3	4	5	6	7	8	9	10	11	12
Design	24	4	4	8	8								
Build	60					8	8	12	12	10	10		
Install & test	16											8	8
Total	100	4	4	8	8	8	8	12	12	10	10	8	8
Cumulative		4	8	16	24	32	40	52	64	74	84	92	100

Amounts are in thousands of Rands.

- Determine budgeted cost by period

FIGURE 7.6 Cumulative Budgeted Cost Curve for the Packaging Machine Project



- Graph the cumulative budgeted cost curve

Packaging machine project determine actual cost

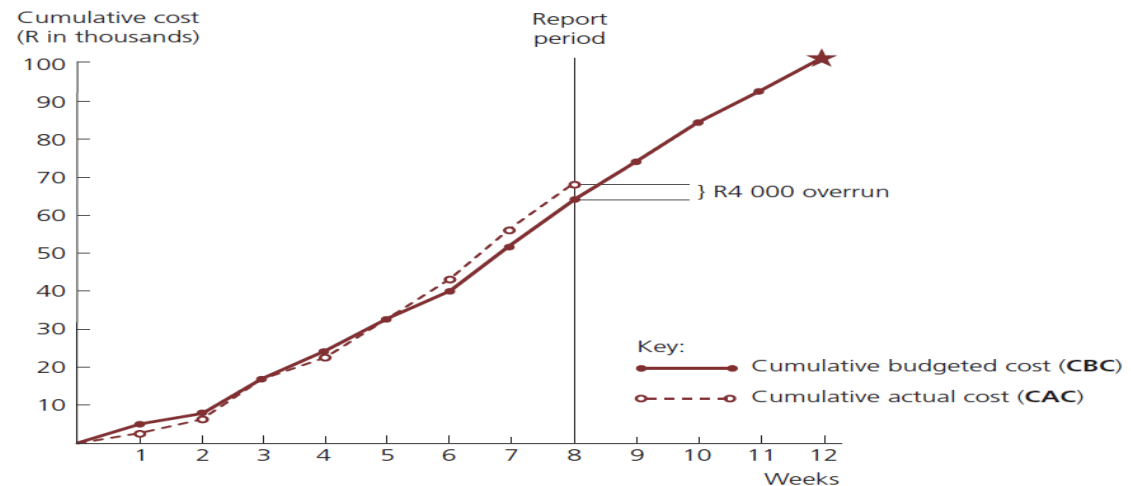
FIGURE 7.7 Actual Cost by Period for the Packaging Machine Project

	Week								Total expended
	1	2	3	4	5	6	7	8	
Design	2	5	9	5	1				22
Build				2	8	10	14	12	46
Install & test									0
Total	2	5	9	7	9	10	14	12	68
Cumulative	2	7	16	23	32	42	56	68	68

Amounts are in thousands of Rands.

- End of Week 8
 - Planned cost = R64,00
 - Actual cost = R68,000

FIGURE 7.8 Cumulative Budgeted and Actual Costs for the Packaging Machine Project



- Compare CAC with CBC

Packaging machine project determine value of work performed

FIGURE 7.10 Cumulative Earned Value by Period for the Packaging Machine Project

		Week							
	TBC	1	2	3	4	5	6	7	8
Design	24	2.4	6	19.2	21.6	24	24	24	24
Build	60				3	9	15	24	30
Install & test	16								
Cumulative	100	2.4	6	19.2	24.6	33	39	48	54

Amounts are in thousands of Rands.

FIGURE 7.9 Cumulative Per Cent Complete by Period for the Packaging Machine Project

		Week							
		1	2	3	4	5	6	7	8
Design		10	25	80	90	100	100	100	100
Build		0	0	0	5	15	25	40	50
Install & test		0	0	0	0	0	0	0	0

Amounts are cumulative per cent complete.

Performing Quantitative Risk Analysis

- Often follows qualitative risk analysis, but both can be done together
- Large, complex projects involving leading edge technologies often require extensive quantitative risk analysis
- Main techniques include:
 - Decision tree analysis
 - Simulation
 - Sensitivity analysis

Decision Trees and Expected Monetary Value (EMV)

- A **decision tree** is a diagramming analysis technique used to help select the best course of action in situations in which future outcomes are uncertain
- **Estimated monetary value (EMV)** is the product of a risk event probability and the risk event's monetary value
- You can draw a decision tree to help find the EMV

Figure 11-7. Expected Monetary Value (EMV) Example

